Panskura Banamali College (Autonomous)

M.Sc. 3rd Semester Examination -2022 (Online Mode)

Subject : Computer Science Paper: COS-301

Paper Name: Advance operating system

F.M.: 40 Time :2 Hours

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1. Answer any two questions: 2×04=8

a)Define OS . Write difference between program and process. 2+2=4

b) Define dispatch latency and semaphore .2+2=4

c)Why OS is called system software ? What is system call? 2+2=4

2. Answer any two questions: 2×08=16

a) Consider the following page-reference string
1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6
How many page faults would occur for the LRU and FIFO replacement algorithms ,if memory consist of three frames.

b) Discus in brief different type of schedulers . Write difference Between pre-emptive and non –pre-emptive scheduling algorithm. 6+2=8

c)Discus about multilevel –feedback queue scheduling algorithm.

Write down the disadvantage of RR scheduling algorithm. 6+2=8

3. Answer any one question 1×16=16

a) Consider the following set of processes , with the length of the CPU –burst time given in milliseconds.

Process	Burst time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The process assumed to have arrived in the order p1 ,p2 , p3 ,p4 , p5 all at time 0.

i) Draw four gantt charts illustrating the execution of these processes using FCFS ,SJF , a non-preemptive priority (a smaller priority number implies a higher priority) and a RR(time quantum=1) scheduling .

ii) What is the turnaround time of each process for each of the scheduling algorithm in part (i) ?

iii) What is the waiting time of each process for each of the scheduling algorithm in part (i) ?

iv) Which of the scheduling in part (i) results in the minimal average waiting time ?

5+5+5+1

b) What is a process switching ? What is context switching ? When do they occur ? What is the difference between the two ?. Why threads are known as tightly coupled ?Describe the differences between symmetric and asymmetric multiprocessing. 3+3+3+1+2+4

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